

ABHINAV PILLAI

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Education

University of Oklahoma

Bachelor of Science - Computer Science

Norman, OK

Aug 2022 - May 2026

Work Experience

University of Oklahoma

Undergraduate Research Assistant

Sep 2025 – May 2026

Norman, OK

- Built an **autonomous humanoid receptionist** on the Unitree G1 EDU Plus using ROS2, enabling real-time navigation, speaker tracking, and human-aware motion.
- Developed an on-device **AI pipeline** on NVIDIA Jetson Orin NX integrating **speech-to-text, LLM-driven dialogue, and retrieval-augmented generation (RAG)** for secure, offline company knowledge delivery.
- Implemented **facial recognition** and **VIP detection** using **depth sensing and vision APIs**, enabling personalized interactions and adaptive response behaviors for high-profile users.

Software Studio @ University of Oklahoma

Startup Mentor and Teaching Assistant

Sep 2025 – May 2026

Norman, OK

- Mentored **40+ students** in building and launching production-ready software products, guiding architecture decisions, feature scoping, and MVP development.
- Led technical workshops on **user stories, feature matrices, prototyping, and system design** to help teams translate ideas into scalable software solutions.
- Advised founders on **product strategy, technical roadmapping, and go-to-market preparation**, supporting teams through live investor pitch cycles.

Vast Vertex Media

Full-stack Software Engineer Intern

Sep 2024 – Apr 2025

Norman, OK

- Developed a scalable **full-stack** mobile application using **React Native, TypeScript, PostgreSQL, Azure, Power BI, and REST APIs**, supporting production deployment and analytics dashboards.
- Designed and built various responsive, user-friendly websites for clients using **Next.js, Tailwind CSS, Figma, and REST APIs**

University of Oklahoma

Undergraduate Research Assistant

Aug 2023 – Dec 2024

Norman, OK

- Developed a full-stack **Next.js/Typescript based** platform, ONEngineering, that allowed for over **400+** active OU Graduate and Alumni Engineering students connect with university contacts and industry professionals
- Built **Python and C++** data ingestion pipelines leveraging RSS feeds to group student and alumni records, reducing recommendation deployment time by **50%**.

UFLIP

Software Engineer Intern

Mar 2023 – Nov 2023

Oklahoma City, OK

- Utilized **CSS, SASS, Node.js, JS, HTML, and Figma** to create a responsive front-end used by 3000+ users
- Improved website startup times by **60%** by replacing synchronous code with **asynchronous** functions and **API calls**

Projects

Route Runners | [Source Code](#) | [Blog](#)

Next.js | TypeScript | Python/Flask

- **Led the development** of a full-stack web application for **first responders**, utilizing **NextJS** and **TypeScript** for the frontend, and **Python + Flask** as the backend
- Implemented a **CI/CD pipeline** to convert drone footage into 3D models enabling **WebAR-based augmented reality** escape plans for first responders and citizens

Technical Skills

Languages: HTML, CSS, JavaScript, Typescript, SQL, NOSQL, Python, Java, C++, C#

Developer Tools: Vercel, Postman, Power BI, Git/Github, Azure, npm, AWS EC2, MS SSMS

Libraries/Frameworks: React, ReactNative, Vue, Angular, Docker, NextJS, NodeJS, .NET